studied from GB sir & dream Maths

— MATRICES

Examples of Matrices

1. $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 6 & 8 \end{bmatrix}$ — order 2×3
2. $\begin{bmatrix} 1 & 5 & 4 \\ 7 & 8 & 9 \\ 3 & 9 & 7 \end{bmatrix}$ — order 3×3
3. $\begin{bmatrix} 1 \\ 6 \\ 7 \\ 8 \end{bmatrix}$ — order 4×1
4. $\begin{bmatrix} x^2+1 & 4x-2 \\ x^3 & x-3 \end{bmatrix}$ — order 2×2

Order of a Matrix

The order of a matrix is defined as:

No. of Rows × No. of Cols.

Examples:

- $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 6 & 8 \end{bmatrix} \rightarrow 2 \times 3$
- $\begin{bmatrix} 1 & 5 & 4 \\ 7 & 8 & 9 \\ 3 & 9 & 7 \end{bmatrix} \rightarrow 3 \times 3$
- $\begin{bmatrix} 1 \\ 6 \\ 7 \\ 8 \end{bmatrix} \rightarrow 4 \times 1$
- $\begin{bmatrix} x^2+1 & 4x-2 \\ x^3 & x-3 \end{bmatrix} \rightarrow 2 \times 2$

Elements of a Matrix

Let

$$A = \begin{bmatrix} 1 & 5 & 4 \\ 7 & 8 & 9 \\ 3 & 9 & 7 \end{bmatrix}.$$

The elements of A are denoted by a_{ij}, where i = row number and j = column number.

$$\begin{array}{lll} a_{11} = 1 & a_{12} = 5 & a_{13} = 4 \\ a_{21} = 7 & a_{22} = 8 & a_{23} = 9 \\ a_{31} = 3 & a_{32} = 9 & a_{33} = 7 \end{array}$$

— Construct a Matrix

Question: Construct a 3×4 matrix $A = [a_{ij}]$, whose element a_{ij} is given by

$$a_{ij} = \frac{(i+j)^2}{2}, \qquad 1 \le i \le 3, \quad 1 \le j \le 4.$$

Solution: We have

$$a_{ij} = \frac{(i+j)^2}{2}, \qquad 1 \le i \le 3, \quad 1 \le j \le 4.$$

To find a_{11}, put $i = 1, j = 1$ in a_{ij}:

$$a_{11} = \frac{(1+1)^2}{2} = \frac{2^2}{2} = \frac{4}{2} = 2.$$

To find a_{12}, put $i = 1, j = 2$ in a_{ij}:

$$a_{12} = \frac{(1+2)^2}{2} = \frac{3^2}{2} = \frac{9}{2}.$$

To find a_{13}, put $i = 1, j = 3$ in a_{ij}:

$$a_{13} = \frac{(1+3)^2}{2} = \frac{4^2}{2} = \frac{16}{2} = 8.$$

To find a_{14}, put $i = 1, j = 4$ in a_{ij}:

$$a_{14} = \frac{(1+4)^2}{2} = \frac{5^2}{2} = \frac{25}{2}.$$

To find a_{21}, put $i = 2, j = 1$ in a_{ij}:

$$a_{21} = \frac{(2+1)^2}{2} = \frac{3^2}{2} = \frac{9}{2}.$$

To find a_{22}, put $i = 2, j = 2$ in a_{ij}:

$$a_{22} = \frac{(2+2)^2}{2} = \frac{4^2}{2} = \frac{16}{2} = 8.$$

To find a_{23}, put $i = 2, j = 3$ in a_{ij}:

$$a_{23} = \frac{(2+3)^2}{2} = \frac{5^2}{2} = \frac{25}{2}.$$

To find a_{24}, put $i = 2, j = 4$ in a_{ij}:

$$a_{24} = \frac{(2+4)^2}{2} = \frac{6^2}{2} = \frac{36}{2} = 18.$$

To find a_{31}, put $i = 3, j = 1$ in a_{ij}:

$$a_{31} = \frac{(3+1)^2}{2} = \frac{4^2}{2} = \frac{16}{2} = 8.$$

To find a_{32}, put $i = 3, j = 2$ in a_{ij}:

$$a_{32} = \frac{(3+2)^2}{2} = \frac{5^2}{2} = \frac{25}{2}.$$

To find a_{33}, put $i = 3, j = 3$ in a_{ij}:

$$a_{33} = \frac{(3+3)^2}{2} = \frac{6^2}{2} = \frac{36}{2} = 18.$$

To find a_{34}, put $i = 3, j = 4$ in a_{ij}:

$$a_{34} = \frac{(3+4)^2}{2} = \frac{7^2}{2} = \frac{49}{2}.$$

Hence,

$$A = \begin{bmatrix} 2 & \frac{9}{2} & 8 & \frac{25}{2} \\ \frac{9}{2} & 8 & \frac{25}{2} & 18 \\ 8 & \frac{25}{2} & 18 & \frac{49}{2} \end{bmatrix}.$$

— Types of Matrices

S. No.	Type of matrix	Definition	Example shown in the source
1	Square Matrix	A matrix is called a square matrix if the number of rows and columns in the matrix are equal.	$\begin{bmatrix}1 & 4 & 7\\2 & 5 & 2\\3 & 6 & 1\end{bmatrix}$, order 3×3
2	Row Matrix	A matrix is called a row matrix if it contains only one row.	$\begin{bmatrix}1 & 5 & 4 & 3\end{bmatrix}$, order 1×4
3	Column Matrix	A matrix is called a column matrix if it contains only one column.	$\begin{bmatrix}1\\4\\2\end{bmatrix}$, order 3×1
4	Zero (or Null) Matrix	A matrix is called a zero matrix if all its elements are zero. It is denoted by '0'.	$\begin{bmatrix}0 & 0\\0 & 0\end{bmatrix}$, order 2×2
5	Diagonal Matrix	A square matrix is called a diagonal matrix if all its non-diagonal elements are zero.	$\begin{bmatrix}1 & 0 & 0\\0 & 3 & 0\\0 & 0 & 6\end{bmatrix}$, order 3×3
6	Scalar Matrix	A square matrix is called a scalar matrix if all its non-diagonal elements are zero and all diagonal elements are equal.	$\begin{bmatrix}4 & 0 & 0\\0 & 4 & 0\\0 & 0 & 4\end{bmatrix}$, order 3×3
7	Unit Matrix or Identity Matrix	A square matrix is called a unit matrix if all its non-diagonal elements are zero and all diagonal elements are unity.	$\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$, order 3×3
8	Triangular Matrices	A square matrix is called a triangular matrix if all its elements either below or above the principal diagonal are zero.	Upper and lower triangular examples below.

(i) Upper Triangular Matrix: If all elements below the principal diagonal are zero.

Example: $\begin{bmatrix}1 & 2 & 3\\0 & 4 & 5\\0 & 0 & 7\end{bmatrix}$, order 3×3.

(ii) Lower Triangular Matrix: If all elements above the principal diagonal are zero.

Example: $\begin{bmatrix}1 & 0 & 0\\4 & 2 & 0\\5 & 6 & 3\end{bmatrix}$, order 3×3.

— Comparable and Equal Matrices

9. Comparable Matrices

Two matrices are called comparable if they are of the same order (i.e., they have the same number of rows and the same number of columns).

Example: Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \qquad B = \begin{bmatrix} 7 & 8 & 9 \\ 1 & 2 & 3 \end{bmatrix}.$$

A and B are comparable. (Both are of order 2×3.)

10. Equal Matrices

Two matrices are called equal matrices if they are of the same order and their corresponding elements are equal.

Example: Let

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \end{bmatrix}.$$

A and B are equal matrices.

— Equality of Matrices: Finding x, y, z

Question 1: Find the values of x, y, z if

$$\begin{bmatrix} 2x + y & x - y \\ x - z & x + y + z \end{bmatrix} = \begin{bmatrix} 10 & -1 \\ 2 & 8 \end{bmatrix}.$$

Solution: By equality of matrices, corresponding elements are equal.

$$\begin{aligned} 2x + y &= 10 \qquad (1) \\ x - y &= -1 \qquad (2) \\ x - z &= 2 \qquad (3) \\ x + y + z &= 8 \qquad (4) \end{aligned}$$

$(1) + (2)$:

$$(2x + y) + (x - y) = 10 + (-1), \qquad 3x = 9, \qquad x = 3.$$

From (2):

$$x - y = -1 \Rightarrow 3 - y = -1 \Rightarrow -y = -4 \Rightarrow y = 4.$$

From (3):

$$x - z = 2 \Rightarrow 3 - z = 2 \Rightarrow -z = -1 \Rightarrow z = 1.$$

Therefore, $x = 3,\ y = 4,\ z = 1$.

Question 2

If

$$\begin{bmatrix} 2x+1 & 2y \\ 0 & y^2+1 \end{bmatrix} = \begin{bmatrix} x+3 & 10 \\ 0 & 26 \end{bmatrix},$$

find the value of $x + y$.

Solution: By equality of matrices,

$$2x + 1 = x + 3 \quad (1), \qquad 2y = 10 \quad (2), \qquad y^2 + 1 = 26 \quad (3).$$

From (1): $2x + 1 = x + 3 \Rightarrow 2x - x = 3 - 1 \Rightarrow x = 2$.

From (2): $2y = 10 \Rightarrow y = 10/2 \Rightarrow y = 5$.

From (3): $y^2 + 1 = 26 \Rightarrow y^2 = 25 \Rightarrow y = \pm 5$. But from (2), $y = 5$.

Therefore, $x + y = 2 + 5 = 7$.

— Equality of Matrices: Finding x, y

Question: For what values of x and y are the following matrices equal?

$$\begin{bmatrix} 2x+1 & 3y \\ 0 & y^2-5y \end{bmatrix}, \qquad \begin{bmatrix} x+3 & y^2+2 \\ 0 & -6 \end{bmatrix}.$$

Solution: For two matrices to be equal, their corresponding elements must be equal. Equating corresponding elements, we get:

$$\begin{aligned} 2x + 1 &= x + 3 && (1) \\ 3y &= y^2 + 2 && (2) \\ 0 &= 0 && (3) \\ y^2 - 5y &= -6 && (4) \end{aligned}$$

From (1):

$$2x + 1 = x + 3 \Rightarrow 2x - x = 3 - 1 \Rightarrow x = 2.$$

From (2):

$$3y = y^2 + 2 \Rightarrow y^2 - 3y + 2 = 0 \Rightarrow (y-1)(y-2) = 0,$$

so $y = 1$ or $y = 2$.

From (4):

$$y^2 - 5y = -6 \Rightarrow y^2 - 5y + 6 = 0 \Rightarrow (y-2)(y-3) = 0,$$

so $y = 2$ or $y = 3$.

From both (2) and (4), the common value of y is $y = 2$.

Therefore, $x = 2,\ y = 2$.

— Operations on Matrices

1. Addition of Matrices

Matrices should have same order.

$$A = \begin{bmatrix} 2 & 3 & 6 \\ 4 & 0 & 9 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 6 & 8 \\ 7 & 0 & 8 \end{bmatrix}.$$

$$A + B = \begin{bmatrix} 2+0 & 3+6 & 6+8 \\ 4+5 & 0+0 & 9+8 \end{bmatrix} = \begin{bmatrix} 2 & 9 & 14 \\ 11 & 0 & 17 \end{bmatrix}.$$

2. Subtraction of Matrices

Matrices should have same order.

$$A - B = \begin{bmatrix} 2-0 & 3-6 & 6-8 \\ 4-7 & 0-0 & 9-8 \end{bmatrix} = \begin{bmatrix} 2 & -3 & -2 \\ -3 & 0 & 1 \end{bmatrix}.$$

3. Scalar Multiplication of Matrix

$$A = \begin{bmatrix} 4 & 7 & 3 \\ 2 & 1 & 2 \end{bmatrix}, \qquad 3A = \begin{bmatrix} 3\times 4 & 3\times 7 & 3\times 3 \\ 3\times 2 & 3\times 1 & 3\times 2 \end{bmatrix} = \begin{bmatrix} 12 & 21 & 9 \\ 6 & 3 & 6 \end{bmatrix}.$$

Another example shown:

$$B = \begin{bmatrix} 2 & 6 & 4 \\ 8 & 12 & 8 \end{bmatrix}, \qquad \frac{1}{2}B = \begin{bmatrix} 1 & 3 & 2 \\ 4 & 6 & 4 \end{bmatrix}.$$

Note: In addition and subtraction, order must be same. In scalar multiplication, any scalar can be used.

— Solve the Matrix Equation

Question: Solve the matrix equation

$$\begin{bmatrix} x^2 \\ y^2 \end{bmatrix} - 3 \begin{bmatrix} x \\ 2y \end{bmatrix} = \begin{bmatrix} -2 \\ 9 \end{bmatrix}.$$

Solution: We have

$$\begin{bmatrix} x^2 \\ y^2 \end{bmatrix} - 3 \begin{bmatrix} x \\ 2y \end{bmatrix} = \begin{bmatrix} -2 \\ 9 \end{bmatrix} \Rightarrow \begin{bmatrix} x^2 \\ y^2 \end{bmatrix} - \begin{bmatrix} 3x \\ 6y \end{bmatrix} = \begin{bmatrix} -2 \\ 9 \end{bmatrix}.$$

$$\begin{bmatrix} x^2 - 3x \\ y^2 - 6y \end{bmatrix} = \begin{bmatrix} -2 \\ 9 \end{bmatrix}.$$

Thus,

$$x^2 - 3x = -2 \quad (1), \qquad y^2 - 6y = 9 \quad (2).$$

From (1):

$$x^2 - 3x + 2 = 0 \Rightarrow (x-1)(x-2) = 0 \Rightarrow x = 1, 2.$$

From (2):

$$y^2 - 6y - 9 = 0 \Rightarrow y^2 - 6y + 9 = 0 - 9 + 9 \Rightarrow (y-3)^2 = 9.$$

$$y - 3 = \pm 3 \Rightarrow y = 3 \pm 3 \Rightarrow y = 6, 0.$$

Therefore, $x = 1, 2$ and $y = 6, 0$.

— Find Matrices X and Y

Question: Find X and Y if

$$2X + Y = \begin{bmatrix} 4 & 4 & 7 \\ 7 & 3 & 4 \end{bmatrix} \quad \text{and} \quad X - 2Y = \begin{bmatrix} -3 & 2 & 1 \\ 1 & -1 & 2 \end{bmatrix}.$$

Solution: We have

$$2X + Y = \begin{bmatrix} 4 & 4 & 7 \\ 7 & 3 & 4 \end{bmatrix} \quad (1)$$

and

$$X - 2Y = \begin{bmatrix} -3 & 2 & 1 \\ 1 & -1 & 2 \end{bmatrix} \quad (2).$$

$(1) \times 2$:

$$4X + 2Y = 2\begin{bmatrix} 4 & 4 & 7 \\ 7 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 8 & 8 & 14 \\ 14 & 6 & 8 \end{bmatrix} \quad (3).$$

$(2) + (3)$:

$$5X = \begin{bmatrix} -3 & 2 & 1 \\ 1 & -1 & 2 \end{bmatrix} + \begin{bmatrix} 8 & 8 & 14 \\ 14 & 6 & 8 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 15 & 5 & 10 \end{bmatrix}.$$

$$X = \frac{1}{5}\begin{bmatrix} 5 & 10 & 15 \\ 15 & 5 & 10 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix}.$$

Putting the value of X in (1), we get

$$\begin{aligned} Y &= \begin{bmatrix} 4 & 4 & 7 \\ 7 & 3 & 4 \end{bmatrix} - 2X \\ &= \begin{bmatrix} 4 & 4 & 7 \\ 7 & 3 & 4 \end{bmatrix} - 2\begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 4 & 7 \\ 7 & 3 & 4 \end{bmatrix} - \begin{bmatrix} 2 & 4 & 6 \\ 6 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}. \end{aligned}$$

Therefore,

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix}, \qquad Y = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

— Multiplication of Matrices

Let A be of order $m_1 \times n_1$ and B be of order $m_2 \times n_2$. For multiplication, n_1 should be the same as m_2.

Multiplication AB possible if: No. of col in A = No. of Row B.

(a)

$$A = \begin{bmatrix} 4 & 7 \\ 6 & 2 \end{bmatrix}_{2\times 2}, \qquad B = \begin{bmatrix} 1 & 8 \\ 3 & 7 \end{bmatrix}_{2\times 2}.$$

$$AB = \begin{bmatrix} 4\times 1 + 7\times 3 & 4\times 8 + 7\times 7 \\ 6\times 1 + 2\times 3 & 6\times 8 + 2\times 7 \end{bmatrix} = \begin{bmatrix} 4+21 & 32+49 \\ 6+6 & 48+14 \end{bmatrix} = \begin{bmatrix} 25 & 81 \\ 12 & 62 \end{bmatrix}.$$

(b)

$$A = \begin{bmatrix} 1 \\ 1 \end{bmatrix}_{2\times 1}, \qquad B = \begin{bmatrix} 2 \\ 4 \end{bmatrix}_{2\times 1}.$$

No. of col of A $\neq$ No. of Row B. **AB not possible.**

(c)

$$A = \begin{bmatrix} 7 & 8 \\ 9 & 8 \\ 6 & 5 \end{bmatrix}_{3\times 2}, \qquad B = \begin{bmatrix} 1 & 5 & 7 \\ 0 & 6 & 8 \end{bmatrix}_{2\times 3}.$$

$$AB = \begin{bmatrix} 7\times 1 + 8\times 0 & 7\times 5 + 8\times 6 & 7\times 7 + 8\times 8 \\ 9\times 1 + 8\times 0 & 9\times 5 + 8\times 6 & 9\times 7 + 8\times 8 \\ 6\times 1 + 5\times 0 & 6\times 5 + 5\times 6 & 6\times 7 + 5\times 8 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 35+48 & 49+64 \\ 9 & 45+48 & 63+64 \\ 6 & 30+30 & 42+40 \end{bmatrix} = \begin{bmatrix} 7 & 83 & 113 \\ 9 & 93 & 127 \\ 6 & 60 & 82 \end{bmatrix}.$$

Note: It is not necessary $AB = BA$.

For the example at the bottom of the page, A is 2×3 and B is 3×4; AB is 2×4, while BA is not defined.

Points to Remember:

- In AB, the number of columns of A must be equal to the number of rows of B.
- In BA, the number of columns of B must be equal to the number of rows of A.

— Positive Integral Powers of Matrices

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}.$$

$$A^2 = A \cdot A = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}.$$

$$= \begin{bmatrix} 2\times 2+(-1)\times 3 & 2\times(-1)+(-1)\times 2 \\ 3\times 2+2\times 3 & 3\times(-1)+2\times 2 \end{bmatrix} = \begin{bmatrix} 4-3 & -2-2 \\ 6+6 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ 12 & 1 \end{bmatrix}.$$

Therefore,

$$A^2 = \begin{bmatrix} 1 & -4 \\ 12 & 1 \end{bmatrix}.$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 1 & -4 \\ 12 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 2-12 & -1-8 \\ 24+3 & -12+2 \end{bmatrix} = \begin{bmatrix} -10 & -9 \\ 27 & -10 \end{bmatrix}.$$

Therefore,

$$A^3 = \begin{bmatrix} -10 & -9 \\ 27 & -10 \end{bmatrix}.$$

$$\boxed{A^3 = A^2 \cdot A \quad \text{or} \quad A \cdot A^2.}$$

— Matrix Polynomial and Higher Powers

Question: Let

$$A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \quad \text{and} \quad f(x) = x^2 - 4x + 7,$$

show that $f(A) = 0$. Using this result, find A^5.

Solution: We have

$$A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}.$$

$f(x) = x^2 - 4x + 7$ implies $f(A) = A^2 - 4A + 7I_2$.

$$f(A) = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}^2 - 4\begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} + 7\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

$$= \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} - \begin{bmatrix} 8 & 12 \\ -4 & 8 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}.$$

$$= \begin{bmatrix} 4-3 & 6+6 \\ -2-2 & -3+4 \end{bmatrix} - \begin{bmatrix} 8 & 12 \\ -4 & 8 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix} - \begin{bmatrix} 8 & 12 \\ -4 & 8 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}.$$

$$= \begin{bmatrix} 1-8+7 & 12-12+0 \\ -4+4+0 & 1-8+7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0.$$

Therefore, $f(A) = 0$, i.e.

$$A^2 - 4A + 7I_2 = 0, \qquad \text{i.e. } A^2 = 4A - 7I_2.$$

$$\begin{aligned} A^3 &= A \cdot A^2 = A(4A - 7I_2) = 4A^2 - 7A \\ &= 4(4A - 7I_2) - 7A = 9A - 28I_2, \\ A^4 &= AA^3 = A(9A - 28I_2) = 9A^2 - 28A \\ &= 9(4A - 7I_2) - 28A = 8A - 63I_2, \\ A^5 &= AA^4 = A(8A - 63I_2) = 8A^2 - 63A \\ &= 8(4A - 7I_2) - 63A = -31A - 56I_2. \end{aligned}$$

Therefore,

$$A^5 = -31 \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} - 56 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -62 & -93 \\ 31 & -62 \end{bmatrix} - \begin{bmatrix} 56 & 0 \\ 0 & 56 \end{bmatrix} = \begin{bmatrix} -118 & -93 \\ 31 & -118 \end{bmatrix}.$$

— Special Square Matrices

Note: A square matrix A is said to be an **idempotent matrix** if $A^2 = A$.

A square matrix A is said to be an **involutory matrix** if $A^2 = I$.

A square matrix A is said to be a **nilpotent matrix** if there exists $m \in \mathbb{N}$ such that $A^m = 0$.

If m is the least natural number such that $A^m = 0$, then m is called the **index of nilpotent matrix** A.

— Matrix Power Example and Idempotent Matrix

Question 1

If

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 4 \\ -1 & 7 \end{bmatrix},$$

find $3A^2 - 2B$.

Solution:

$$A^2 = AA = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 4-3 & -2-2 \\ 6+6 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ 12 & 1 \end{bmatrix}.$$

$$\begin{aligned} 3A^2 - 2B &= 3\begin{bmatrix} 1 & -4 \\ 12 & 1 \end{bmatrix} - 2\begin{bmatrix} 0 & 4 \\ -1 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 3 & -12 \\ 36 & 3 \end{bmatrix} - \begin{bmatrix} 0 & 8 \\ -2 & 14 \end{bmatrix} \\ &= \begin{bmatrix} 3-0 & -12-8 \\ 36+2 & 3-14 \end{bmatrix} = \begin{bmatrix} 3 & -20 \\ 38 & -11 \end{bmatrix}. \end{aligned}$$

Question 2

Show that the matrix

$$A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$$

is idempotent.

Solution: Matrix A is idempotent if $A^2 = A$.

$$A^2 = AA = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}.$$

$$= \begin{bmatrix} 4+2-4 & -4-6+8 & -8-8+12 \\ -2-3+4 & 2+9-8 & 4+12-12 \\ 2+2-3 & -2-6+6 & -4-8+9 \end{bmatrix} = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = A.$$

Therefore, matrix A is idempotent.

— Transpose of a Matrix

Transpose of a Matrix

Example (i):

$$A = \begin{bmatrix} 2 & 3 & 6 & 8 \\ 5 & -3 & -7 & 4 \\ 9 & 8 & 2 & 1 \end{bmatrix}_{3\times 4} \quad \Rightarrow \quad A' = \begin{bmatrix} 2 & 5 & 9 \\ 3 & -3 & 8 \\ 6 & -7 & 2 \\ 8 & 4 & 1 \end{bmatrix}_{4\times 3}.$$

Example (ii):

$$A = \begin{bmatrix} 5 \\ 9 \\ 0 \end{bmatrix}_{3\times 1}, \qquad A' = [5 \quad 9 \quad 0]_{1\times 3}.$$

Properties of Transpose

$$(A')' = A,$$

$$(A+B)' = A' + B',$$

$$(kA)' = kA',$$

$$(AB)' = B'A'.$$

— Orthogonal Matrix

A square matrix A is said to be an **orthogonal matrix** if

$$AA' = I = A'A.$$

Let

$$A = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \qquad A' = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Then

$$AA' = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1+1 & 1-1 \\ 1-1 & 1+1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = I.$$

And

$$A'A = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1+1 & -1+1 \\ -1+1 & 1+1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = I.$$

Therefore, $AA' = I = A'A$, and A is an orthogonal matrix.

— Symmetric Matrix and Skew-Symmetric Matrix

Symmetric Matrix

Property: A square matrix A is said to be symmetric if $A' = A$.

Example:

$$A = \begin{bmatrix} 3 & 4 & 7 \\ 4 & 9 & 8 \\ 7 & 8 & 11 \end{bmatrix}.$$

$$A' = \begin{bmatrix} 3 & 4 & 7 \\ 4 & 9 & 8 \\ 7 & 8 & 11 \end{bmatrix} = A.$$

Therefore, $A' = A$, and A is symmetric.

Skew-Symmetric Matrix

Property: A square matrix A is said to be skew-symmetric if $A' = -A$.

Example:

$$A = \begin{bmatrix} 0 & 9 & -6 \\ -9 & 0 & 8 \\ 6 & -8 & 0 \end{bmatrix}.$$

$$A' = \begin{bmatrix} 0 & -9 & 6 \\ 9 & 0 & -8 \\ -6 & 8 & 0 \end{bmatrix} = - \begin{bmatrix} 0 & 9 & -6 \\ -9 & 0 & 8 \\ 6 & -8 & 0 \end{bmatrix} = -A.$$

Therefore, $A' = -A$, and A is skew-symmetric.

In General

- If A is symmetric, then $a_{ij} = a_{ji}$ for all i, j.
- If A is skew-symmetric, then $a_{ij} = -a_{ji}$ for all i, j.

— Sums and Differences with the Transpose

Question: If

$$A = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix},$$

show that:

(i) $A + A'$ is symmetric.
(ii) $A - A'$ is skew-symmetric.

Solution: We have

$$A = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix}, \qquad A' = \begin{bmatrix} 2 & 5 \\ 4 & 6 \end{bmatrix}.$$

(i)

$$A + A' = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} 2 & 5 \\ 4 & 6 \end{bmatrix} = \begin{bmatrix} 2+2 & 4+5 \\ 5+4 & 6+6 \end{bmatrix} = \begin{bmatrix} 4 & 9 \\ 9 & 12 \end{bmatrix}.$$

$(A + A')' = A + A'$. Thus, $A + A'$ is symmetric.

(ii)

$$A - A' = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix} - \begin{bmatrix} 2 & 5 \\ 4 & 6 \end{bmatrix} = \begin{bmatrix} 2-2 & 4-5 \\ 5-4 & 6-6 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

$(A - A')' = -(A - A')$. Thus, $A - A'$ is skew-symmetric.

— Express a Matrix as a Sum of Symmetric and Skew-Symmetric Matrices

Question: Express

$$\begin{bmatrix} 4 & 3 & 7 \\ 6 & 5 & -8 \\ 1 & 2 & 6 \end{bmatrix}$$

as the sum of a symmetric matrix and a skew-symmetric matrix.

Solution: Let

$$A = \begin{bmatrix} 4 & 3 & 7 \\ 6 & 5 & -8 \\ 1 & 2 & 6 \end{bmatrix}, \qquad A' = \begin{bmatrix} 4 & 6 & 1 \\ 3 & 5 & 2 \\ 7 & -8 & 6 \end{bmatrix}.$$

$$A + A' = \begin{bmatrix} 8 & 9 & 8 \\ 9 & 10 & -6 \\ 8 & -6 & 12 \end{bmatrix}, \qquad A - A' = \begin{bmatrix} 0 & -3 & 6 \\ 3 & 0 & -10 \\ -6 & 10 & 0 \end{bmatrix}.$$

Now,

$$\frac{1}{2}(A+A') = \begin{bmatrix} 4 & \frac{9}{2} & 4 \\ \frac{9}{2} & 5 & -3 \\ 4 & -3 & 6 \end{bmatrix},$$

which is symmetric, and

$$\frac{1}{2}(A-A') = \begin{bmatrix} 0 & -\frac{3}{2} & 3 \\ \frac{3}{2} & 0 & -5 \\ -3 & 5 & 0 \end{bmatrix},$$

which is skew-symmetric.

Therefore,

$$A = \frac{1}{2}(A+A') + \frac{1}{2}(A-A')$$

$$= \begin{bmatrix} 4 & \frac{9}{2} & 4 \\ \frac{9}{2} & 5 & -3 \\ 4 & -3 & 6 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{3}{2} & 3 \\ \frac{3}{2} & 0 & -5 \\ -3 & 5 & 0 \end{bmatrix}.$$

— Determinants

(i)

$$A = \begin{bmatrix} 2 & -3 \\ 4 & 9 \end{bmatrix}.$$

$$|A| = \begin{vmatrix} 2 & -3 \\ 4 & 9 \end{vmatrix} = 18 - (-12) = 18 + 12 = 30.$$

(ii)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}.$$

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 1\begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 2\begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 3\begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix}.$$

$$= 1(45-48) - 2(36-42) + 3(32-35) = (-3) - 2(-6) + 3(-3) = -3 + 12 - 9 = 9 - 9 = 0.$$

— Minors and Co-Factors

Let

$$A = \begin{bmatrix} 4 & 3 & 8 \\ 6 & 7 & 5 \\ 3 & 1 & 2 \end{bmatrix}.$$

Minors

M_{ij} = minor of element a_{ij}.

$$M_{11} = \begin{vmatrix} 7 & 5 \\ 1 & 2 \end{vmatrix} = 14 - 5 = 9.$$

$$M_{12} = \begin{vmatrix} 6 & 5 \\ 3 & 2 \end{vmatrix} = 12 - 15 = -3.$$

$$M_{13} = \begin{vmatrix} 6 & 7 \\ 3 & 1 \end{vmatrix} = 6 - 21 = -15.$$

$$M_{21} = \begin{vmatrix} 3 & 8 \\ 1 & 2 \end{vmatrix} = 6 - 8 = -2.$$

$$M_{22} = \begin{vmatrix} 4 & 8 \\ 3 & 2 \end{vmatrix} = 8 - 24 = -16.$$

$$M_{23} = \begin{vmatrix} 4 & 3 \\ 3 & 1 \end{vmatrix} = 4 - 9 = -5.$$

$$M_{31} = \begin{vmatrix} 3 & 8 \\ 7 & 5 \end{vmatrix} = 15 - 56 = -41.$$

$$M_{32} = \begin{vmatrix} 4 & 8 \\ 6 & 5 \end{vmatrix} = 20 - 48 = -28.$$

$$M_{33} = \begin{vmatrix} 4 & 3 \\ 6 & 7 \end{vmatrix} = 28 - 18 = 10.$$

Matrix of Minors $[M_{ij}]$:

$$[M_{ij}] = \begin{bmatrix} 9 & -3 & -15 \\ -2 & -16 & -5 \\ -41 & -28 & 10 \end{bmatrix}.$$

Cofactors

A_{ij} = co-factor of element $a_{ij} = (-1)^{i+j} M_{ij}$.

$$A_{11} = (-1)^{1+1} M_{11} = (+1)(9) = 9.$$

$$A_{12} = (-1)^{1+2} M_{12} = (-1)(-3) = 3.$$

$$A_{13} = (-1)^{1+3} M_{13} = (+1)(-15) = -15.$$

$$A_{21} = (-1)^{2+1} M_{21} = (-1)(-2) = 2.$$

$$A_{22} = (-1)^{2+2} M_{22} = (+1)(-16) = -16.$$

$$A_{23} = (-1)^{2+3} M_{23} = (-1)(-5) = 5.$$

$$A_{31} = (-1)^{3+1} M_{31} = (+1)(-41) = -41.$$

$$A_{32} = (-1)^{3+2} M_{32} = (-1)(-28) = 28.$$

$$A_{33} = (-1)^{3+3} M_{33} = (+1)(10) = 10.$$

Matrix of Co-factors $[A_{ij}]$:

$$[A_{ij}] = \begin{bmatrix} 9 & 3 & -15 \\ 2 & -16 & 5 \\ -41 & 28 & 10 \end{bmatrix}.$$

— Notes on Determinants

Note 1:

$$\begin{vmatrix} 0 & 1 & 4 \\ 0 & 2 & 5 \\ 0 & 3 & 6 \end{vmatrix} = 0.$$

$$\begin{vmatrix} 0 & 0 & 0 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{vmatrix} = 0.$$

2. The value of determinant of a skew-symmetric matrix of odd order is always zero.

3.

$$a_{11} A_{21} + a_{12} A_{22} + a_{13} A_{23} = 0.$$

The sum of the products of the elements of any row (respectively, column) of a determinant with the cofactors of the corresponding elements of any other row (respectively, column) is zero.

— Sarrus Rule

Sarrus Rule — [only for determinants of order three.]

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 5 & 3 & 7 \\ 1 & 2 & 6 \end{bmatrix}.$$

$$|A| = \begin{vmatrix} 2 & 1 & 3 \\ 5 & 3 & 7 \\ 1 & 2 & 6 \end{vmatrix}.$$

$$= (2)(3)(6) + (1)(7)(1) + (3)(5)(2) - (1)(3)(1) - (2)(7)(2) - (6)(5)(1).$$

$$= 36 + 7 + 30 - 9 - 28 - 30 = 6.$$

Sarrus Rule for 3×3 Determinant

- Repeat the first two columns on the right side.
- Multiply the elements of the three downward diagonals and add them.
- Multiply the elements of the three upward diagonals and subtract them.
- $|A| = (\text{Sum of downward products}) - (\text{Sum of upward products}).$

— Singular and Non-Singular Matrices

Singular Matrix: A square matrix is called a singular matrix if its determinant is zero.

Non-Singular Matrix: A square matrix is called a non-singular matrix if its determinant is not zero.

(i)

$$A = \begin{bmatrix} 4 & 3 \\ 6 & 9 \end{bmatrix}.$$

$$|A| = \begin{vmatrix} 4 & 3 \\ 6 & 9 \end{vmatrix} = 36 - 18 = 18 \neq 0.$$

A is non-singular.

(ii)

$$A = \begin{bmatrix} 1 & 4 & 3 \\ 6 & 8 & -5 \\ 2 & 8 & 6 \end{bmatrix}.$$

$$|A| = 1\begin{vmatrix} 8 & -5 \\ 8 & 6 \end{vmatrix} - 4\begin{vmatrix} 6 & -5 \\ 2 & 6 \end{vmatrix} + 3\begin{vmatrix} 6 & 8 \\ 2 & 8 \end{vmatrix} = 88 - 184 + 96 = 0.$$

A is singular.

— Cramer's Rule

Cramer's Rule: Solution Cases

- If $D \neq 0$: **Unique Solution**

$$x = \frac{D_1}{D}, \quad y = \frac{D_2}{D}, \quad z = \frac{D_3}{D}.$$

- If $D = 0$ and D_1, D_2, D_3 are not all zero: **No Solution**.
- If $D = 0$ and D_1, D_2, D_3 are all zero: **Infinitely many Solutions or no Solution**.

Question

The source heading reads: “Solve the the Using Determinants, solve the following System of equations.” The question is to solve the following system of equations using determinants:

$$x + y + 2z = 0,$$

$$x - 2y + 2z = 0,$$

$$x - z = 0.$$

Solution: The equations are

$$x + y + 2z = 0, \qquad x - 2y + 2z = 0, \qquad x + 0y - z = 0.$$

$$D = \begin{vmatrix} 1 & 1 & 2 \\ 1 & -2 & 2 \\ 1 & 0 & -1 \end{vmatrix} = 1(2 - 0) - 1(-1 - 2) + 2(0 + 2) = 2 + 3 + 4 = 9 \neq 0.$$

Unique solution.

$$D_1 = \begin{vmatrix} 0 & 1 & 2 \\ 0 & -2 & 2 \\ 0 & 0 & -1 \end{vmatrix} = 0, \quad D_2 = \begin{vmatrix} 1 & 0 & 2 \\ 1 & 0 & 2 \\ 1 & 0 & -1 \end{vmatrix} = 0, \quad D_3 = \begin{vmatrix} 1 & 1 & 0 \\ 1 & -2 & 0 \\ 1 & 0 & 0 \end{vmatrix} = 0.$$

$$x = \frac{D_1}{D} = \frac{0}{9} = 0, \quad y = \frac{D_2}{D} = \frac{0}{9} = 0, \quad z = \frac{D_3}{D} = \frac{0}{9} = 0.$$

$$\boxed{x = 0,\ y = 0,\ z = 0.}$$

Unique Solution.

— Properties of the Adjugate

Note: If A is a square matrix of order n, then

$$A(\text{adj } A) = (\text{adj } A)A = |A|I_n.$$

1. $|\text{adj } A| = |A|^{n-1}$.
2. $\text{adj}(\text{adj } A) = |A|^{n-2}A$.
3. $\text{adj } A = (\text{adj } A)'$.

— Inverse of a Matrix

Step 1: Find the value of $|A|$.

Step 2: If $|A| = 0$, then A cannot have its inverse.

Step 3: If $|A| \neq 0$, then A has its inverse and is given by

$$A^{-1} = \frac{\text{adj } A}{|A|}.$$

Step 4: Find cofactors of all elements of A and compute $\text{adj } A$.

Step 5: Compute $\text{adj } A$ (transpose of cofactor matrix).

Step 6: Multiply $\text{adj } A$ by $\frac{1}{|A|}$. That is,

$$A^{-1} = \frac{1}{|A|}(\text{adj } A).$$

Step 7: The result obtained is A^{-1}.

— Find the Inverse and Verify the Result

Question: Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}$$

and verify the result.

Solution: Let

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}.$$

$$|A| = \begin{vmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{vmatrix} = 1(8-6) + 1(0+9) + 2(0-6) = 2 + 9 - 12 = -1 \neq 0.$$

Therefore, A is invertible and

$$A^{-1} = \frac{\operatorname{adj} A}{|A|}.$$

We find $\operatorname{adj} A$

$$A_{11} = +\begin{vmatrix} 2 & -3 \\ -2 & 4 \end{vmatrix} = 2, \quad A_{12} = -\begin{vmatrix} 0 & -3 \\ 3 & 4 \end{vmatrix} = -9, \quad A_{13} = +\begin{vmatrix} 0 & 2 \\ 3 & -2 \end{vmatrix} = -6.$$

$$A_{21} = -\begin{vmatrix} -1 & 2 \\ -2 & 4 \end{vmatrix} = 0, \quad A_{22} = +\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = -2, \quad A_{23} = -\begin{vmatrix} 1 & -1 \\ 3 & -2 \end{vmatrix} = -1.$$

$$A_{31} = +\begin{vmatrix} -1 & 2 \\ 2 & -3 \end{vmatrix} = -1, \quad A_{32} = -\begin{vmatrix} 1 & 2 \\ 0 & -3 \end{vmatrix} = 3, \quad A_{33} = +\begin{vmatrix} 1 & -1 \\ 0 & 2 \end{vmatrix} = 2.$$

Therefore, as shown in the source,

$$\operatorname{adj} A = \begin{bmatrix} 2 & 0 & -1 \\ -9 & -2 & 3 \\ -6 & -1 & 2 \end{bmatrix}.$$

Hence,

$$A^{-1} = \frac{\operatorname{adj} A}{|A|} = \frac{1}{-1}\begin{bmatrix} 2 & 0 & -1 \\ -9 & -2 & 3 \\ -6 & -1 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}.$$

Verification

$$AA^{-1} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix} \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}.$$

$$= \begin{bmatrix} -2-9+12 & 0-2+2 & 1+3-4 \\ 0+18-18 & 0+4-3 & 0-6+6 \\ -6-18+24 & 0-4+4 & -3+6-8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I.$$

Similarly, it can be verified that $A^{-1}A = I$.

— Further Properties of Inverses and Adjugates

Note

1. $(A^{-1})^{-1} = A$.
2. $(A')^{-1} = (A^{-1})'$.
3. $(\operatorname{adj} A)^{-1} = \operatorname{adj}(A^{-1})$.
4. $(AB)^{-1} = B^{-1}A^{-1}$.
5. $\operatorname{adj}(AB) = (\operatorname{adj} B)(\operatorname{adj} A)$.

— Solution of System of Linear Equations by Matrix Method

System: $AX = B$.

Find $|A|$.

- If $|A| \neq 0$: **Unique solution**, $X = A^{-1}B$.
- If $|A| = 0$: find $(\operatorname{adj} A)B$.
  - If $(\operatorname{adj} A)B \neq 0$: **No solution**.
  - If $(\operatorname{adj} A)B = 0$: **Infinitely many solutions or no solution**.

— Solve the Equations by Matrix Method

Question: Solve the equations:

$$x + 2y + z = 7, \qquad x + 3z = 11, \qquad 2x - 3y = 1.$$

Solution: The given equations are

$$x + 2y + z = 7,$$

$$x + 0y + 3z = 11,$$

$$2x - 3y + 0z = 1.$$

Let

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 3 \\ 2 & -3 & 0 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 7 \\ 11 \\ 1 \end{bmatrix}; \qquad AX = B.$$

$$|A| = \begin{vmatrix} 1 & 2 & 1 \\ 1 & 0 & 3 \\ 2 & -3 & 0 \end{vmatrix} = 1(9) - 2(-6) + 1(-3) = 18 \neq 0.$$

Therefore, the system has unique solution $X = A^{-1}B$.

For A, we have

$$\begin{array}{lll} A_{11} = 0 + 9 = 9, & A_{12} = -(0 - 6) = 6, & A_{13} = -3 - 0 = -3, \\ A_{21} = -(0 + 3) = -3, & A_{22} = 0 - 2 = -2, & A_{23} = -(-3 - 4) = 7, \\ A_{31} = 6 - 0 = 6, & A_{32} = -(3 - 1) = -2, & A_{33} = 0 - 2 = -2. \end{array}$$

$$\operatorname{adj} A = \begin{bmatrix} 9 & 6 & -3 \\ -3 & -2 & 7 \\ 6 & -2 & -2 \end{bmatrix}' = \begin{bmatrix} 9 & -3 & 6 \\ 6 & -2 & -2 \\ -3 & 7 & -2 \end{bmatrix}.$$

$$A^{-1} = \frac{\operatorname{adj} A}{|A|} = \frac{1}{18} \begin{bmatrix} 9 & -3 & 6 \\ 6 & -2 & -2 \\ -3 & 7 & -2 \end{bmatrix}.$$

The solution is $X = A^{-1}B$.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{18} \begin{bmatrix} 9 & -3 & 6 \\ 6 & -2 & -2 \\ -3 & 7 & -2 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \\ 1 \end{bmatrix} = \frac{1}{18} \begin{bmatrix} 63 - 33 + 6 \\ 42 - 22 - 2 \\ -21 + 77 - 2 \end{bmatrix} = \frac{1}{18} \begin{bmatrix} 36 \\ 18 \\ 54 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}.$$

Therefore, $x = 2,\ y = 1,\ z = 3.$

— Solve the Equations with Reciprocals by Matrix Method

Question: Solve the equations:

$$\frac{2}{x} + \frac{3}{y} + \frac{10}{z} = 4, \qquad \frac{4}{x} - \frac{6}{y} + \frac{5}{z} = 1, \qquad \frac{6}{x} + \frac{9}{y} - \frac{20}{z} = 2.$$

Solution: The given equations are

$$2\left(\frac{1}{x}\right)+3\left(\frac{1}{y}\right)+10\left(\frac{1}{z}\right)=4 \quad (1)$$

$$4\left(\frac{1}{x}\right)-6\left(\frac{1}{y}\right)+5\left(\frac{1}{z}\right)=1 \quad (2)$$

$$6\left(\frac{1}{x}\right)+9\left(\frac{1}{y}\right)-20\left(\frac{1}{z}\right)=2 \quad (3)$$

Let

$$A=\begin{bmatrix}2 & 3 & 10\\ 4 & -6 & 5\\ 6 & 9 & -20\end{bmatrix},\quad X=\begin{bmatrix}1/x\\ 1/y\\ 1/z\end{bmatrix},\quad B=\begin{bmatrix}4\\ 1\\ 2\end{bmatrix};\qquad AX=B.$$

$$|A|=\begin{vmatrix}2 & 3 & 10\\ 4 & -6 & 5\\ 6 & 9 & -20\end{vmatrix}=2(120+45)-3(-80-30)+10(36+36)$$

$$=2(165)-3(-140)+10(72)=330+330+720=1380\neq 0.$$

Therefore, the system has unique solution $X=A^{-1}B$.

For A, we have

$$\begin{aligned}
&A_{11}=(120+45)=165, && A_{12}=-(-80-30)=110, && A_{13}=(36+36)=72,\\
&A_{21}=-(-60-50)=110, && A_{22}=(-40-60)=-100, && A_{23}=-(-40-36)=76,\\
&A_{31}=(-18+24)=6, && A_{32}=-(-40+20)=20, && A_{33}=(-24-12)=-36.
\end{aligned}$$

$$\operatorname{adj}A=\begin{bmatrix}165 & 110 & 72\\ 110 & -100 & 76\\ 6 & 20 & -36\end{bmatrix}'=\begin{bmatrix}165 & 110 & 6\\ 110 & -100 & 20\\ 72 & 76 & -36\end{bmatrix}.$$

$$A^{-1}=\frac{1}{1380}\begin{bmatrix}165 & 110 & 6\\ 110 & -100 & 20\\ 72 & 76 & -36\end{bmatrix}.$$

The solution is $X=A^{-1}B$.

$$\begin{bmatrix}1/x\\ 1/y\\ 1/z\end{bmatrix}=\frac{1}{1380}\begin{bmatrix}165 & 110 & 6\\ 110 & -100 & 20\\ 72 & 76 & -36\end{bmatrix}\begin{bmatrix}4\\ 1\\ 2\end{bmatrix}=\frac{1}{1380}\begin{bmatrix}660+110+12\\ 440-100+40\\ 288+76-72\end{bmatrix}=\frac{1}{1380}\begin{bmatrix}782\\ 380\\ 292\end{bmatrix}.$$

The source then states:

$$\frac{1}{x}=\frac{782}{1380}=\frac{17}{30},\qquad \frac{1}{y}=\frac{380}{1380}=\frac{19}{69},\qquad \frac{1}{z}=\frac{292}{1380}=\frac{73}{345}.$$

Therefore, the printed answer is

$$\boxed{x=\frac{30}{17},\quad y=\frac{69}{19},\quad z=\frac{345}{73}.}$$

— Reciprocal Equations, Substitution, and Matrix Method

Question: Solve the equations:

$$\frac{2}{x}+\frac{3}{y}+\frac{10}{z}=4,\qquad \frac{4}{x}-\frac{6}{y}+\frac{5}{z}=1,\qquad \frac{6}{x}+\frac{9}{y}-\frac{20}{z}=2.$$

Solution: The given equations are

$$2\left(\frac{1}{x}\right)+3\left(\frac{1}{y}\right)+10\left(\frac{1}{z}\right)=4,$$

$$4\left(\frac{1}{x}\right)-6\left(\frac{1}{y}\right)+5\left(\frac{1}{z}\right)=1,$$

$$6\left(\frac{1}{x}\right)+9\left(\frac{1}{y}\right)-20\left(\frac{1}{z}\right)=2.$$

Let

$$u=\frac{1}{x},\qquad v=\frac{1}{y},\qquad w=\frac{1}{z}.$$

Then the equations become

$$2u+3v+10w=4,$$

$$4u-6v+5w=1,$$

$$6u+9v-20w=2.$$

Let

$$A=\begin{bmatrix}2 & 3 & 10\\ 4 & -6 & 5\\ 6 & 9 & -20\end{bmatrix},\quad X=\begin{bmatrix}u\\ v\\ w\end{bmatrix},\quad B=\begin{bmatrix}4\\ 1\\ 2\end{bmatrix};\qquad AX=B.$$

$$|A| = \begin{vmatrix} 2 & 3 & 10 \\ 4 & -6 & 5 \\ 6 & 9 & -20 \end{vmatrix} = 2(120-45) - 3(-80-30) + 10(36+36)$$

$$= 2(75) - 3(-110) + 10(72) = 150 + 330 + 720 = 1200 \neq 0.$$

Therefore, the system has unique solution $X = A^{-1}B$.

For A, we have

$$\begin{aligned} A_{11} &= 120, & A_{12} &= -80, & A_{13} &= 36, \\ A_{21} &= -45, & A_{22} &= 75, & A_{23} &= -36, \\ A_{31} &= 45, & A_{32} &= -30, & A_{33} &= -18. \end{aligned}$$

$$\operatorname{adj} A = \begin{bmatrix} 120 & -45 & 45 \\ -80 & 75 & -30 \\ 36 & -36 & -18 \end{bmatrix}' = \begin{bmatrix} 120 & -80 & 36 \\ -45 & 75 & -36 \\ 45 & -30 & -18 \end{bmatrix}.$$

$$A^{-1} = \frac{1}{1200} \begin{bmatrix} 120 & -80 & 36 \\ -45 & 75 & -36 \\ 45 & -30 & -18 \end{bmatrix}.$$

The solution is $X = A^{-1}B$.

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \frac{1}{1200} \begin{bmatrix} 120 & -80 & 36 \\ -45 & 75 & -36 \\ 45 & -30 & -18 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} = \frac{1}{1200} \begin{bmatrix} 480 - 80 + 72 \\ -180 + 75 - 72 \\ 180 - 30 - 36 \end{bmatrix} = \frac{1}{1200} \begin{bmatrix} 472 \\ -177 \\ 114 \end{bmatrix}.$$

Therefore,

$$u = \frac{59}{150}, \qquad v = -\frac{59}{400}, \qquad w = \frac{19}{200}.$$

That is,

$$\frac{1}{x} = \frac{59}{150}, \qquad \frac{1}{y} = -\frac{59}{400}, \qquad \frac{1}{z} = \frac{19}{200}.$$

Therefore,

$$\boxed{x = \frac{150}{59}, \quad y = -\frac{400}{59}, \quad z = \frac{200}{19}.}$$

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